Milestone 1: Formula 1 Podium Prediction Companion

Analytical Report

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1 Data Cleaning & Auditing

1.1 Initial EDA and Sentinel Handling

During the initial Exploratory Data Analysis (EDA), the missing data was identified across the CSV files as it was represented by the literal string sentinel `\N`. These placeholders were parsed into standard `NaN` values to properly evaluate missingness and prevent numerical errors.

Also, semantic typing was applied to format the raw data for modeling. Date columns (like race dates and driver dates of birth) were converted into datetime objects to allow for age calculations and chronological sorting. Qualifying lap times (`q1`, `q2`, `q3`) were transformed from string formats (MM:SS.ms) into continuous numeric seconds.

These figures are examples that illustrate how the tables looked like before cleaning:

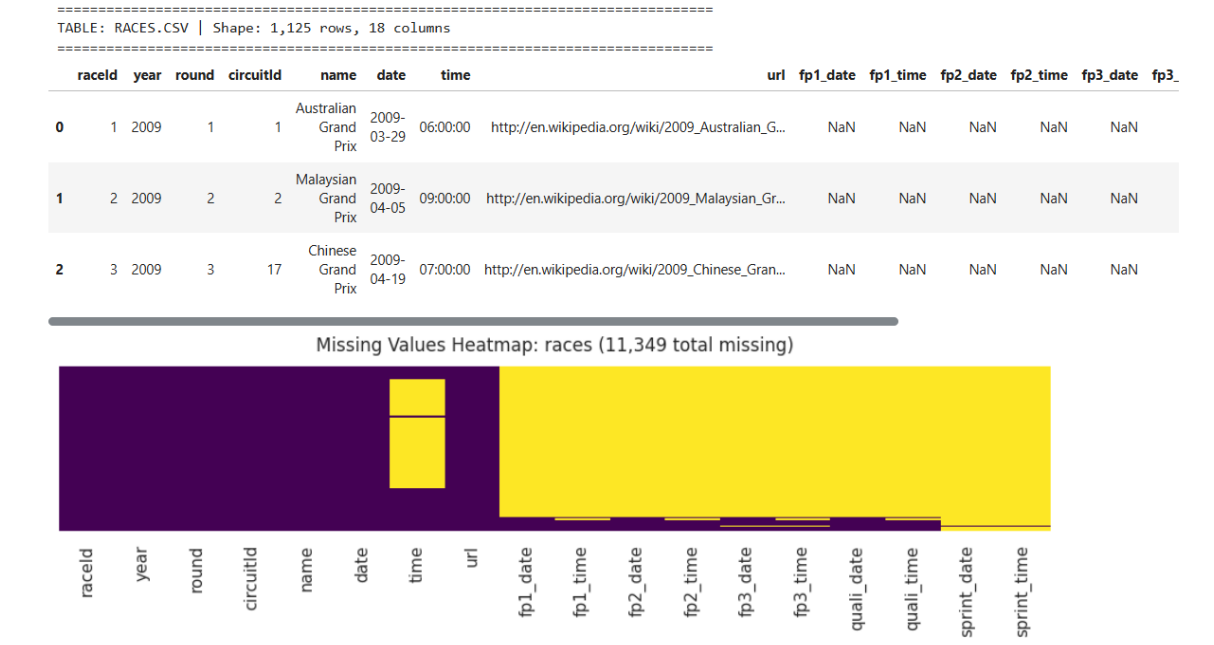


TABLE: RACES.CSV | Shape: 1,125 rows, 18 columns

	raceId	year	round	circuitId	name	date	time	url	fp1_date	fp1_time	fp2_date	fp2_time	fp3_date	fp3_
0	1	2009	1	1	Australian Grand Prix	2009-03-29	06:00:00	http://en.wikipedia.org/wiki/2009_Australian_G...	NaN	NaN	NaN	NaN	NaN	
1	2	2009	2	2	Malaysian Grand Prix	2009-04-05	09:00:00	http://en.wikipedia.org/wiki/2009_Malaysian_Gr...	NaN	NaN	NaN	NaN	NaN	
2	3	2009	3	17	Chinese Grand Prix	2009-04-19	07:00:00	http://en.wikipedia.org/wiki/2009_Chinese_Gran...	NaN	NaN	NaN	NaN	NaN	

Figure 1: The Races Table

```
================================================================================
TABLE: RESULTS.CSV | Shape: 26,759 rows, 18 columns
================================================================================
```

	resultId	raceId	driverId	constructorId	number	grid	position	positionText	positionOrder	points	laps	time	milliseconds	fastestLap	rank	fastestL
0	1	18	1	1	22.0	1	1.0	1	1	10.0	58	1:34:50.616	5690616.0	39.0	2.0	
1	2	18	2	2	3.0	5	2.0	2	2	8.0	58	+5.478	5696094.0	41.0	3.0	
2	3	18	3	3	7.0	7	3.0	3	3	6.0	58	+8.163	5698779.0	41.0	5.0	

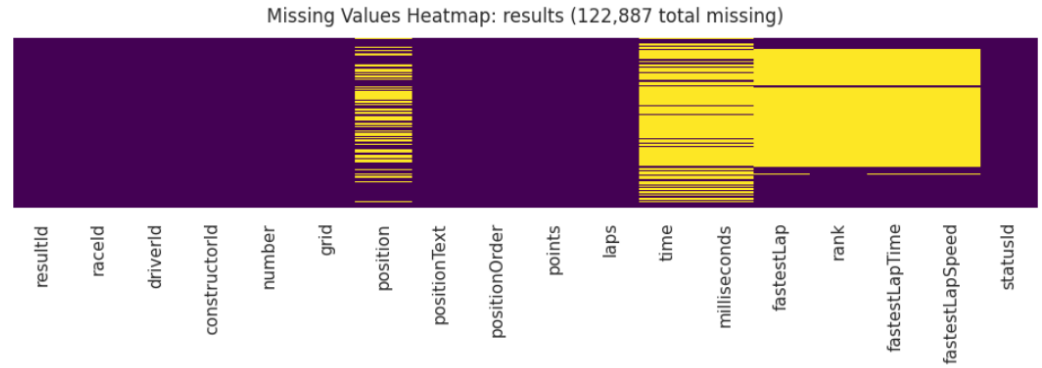


Figure 2: The Results Table

```
================================================================================
TABLE: DRIVERS.CSV | Shape: 861 rows, 9 columns
================================================================================
```

	driverId	driverRef	number	code	forename	surname	dob	nationality	url
0	1	hamilton	44.0	HAM	Lewis	Hamilton	1985-01-07	British	http://en.wikipedia.org/wiki/Lewis_Hamilton
1	2	heidfeld	NaN	HEI	Nick	Heidfeld	1977-05-10	German	http://en.wikipedia.org/wiki/Nick_Heidfeld
2	3	rosberg	6.0	ROS	Nico	Rosberg	1985-06-27	German	http://en.wikipedia.org/wiki/Nico_Rosberg

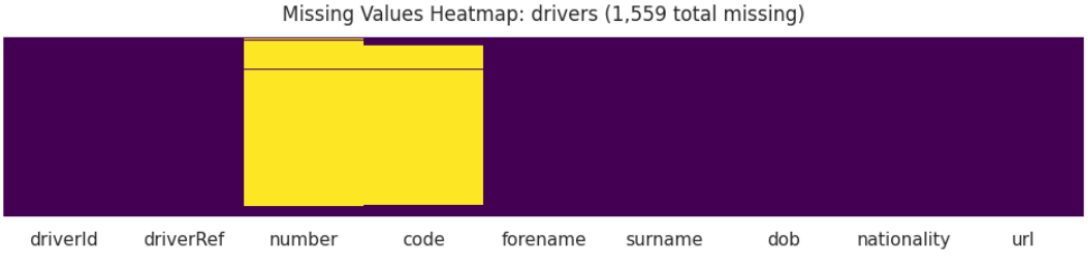


Figure 3: The Drivers Table

TABLE: QUALIFYING.CSV | Shape: 10,494 rows, 9 columns

	qualifyId	raceId	driverId	constructorId	number	position	q1	q2	q3
0	1	18	1	1	22	1	1:26.572	1:25.187	1:26.714
1	2	18	9	2	4	2	1:26.103	1:25.315	1:26.869
2	3	18	5	1	23	3	1:25.664	1:25.452	1:27.079

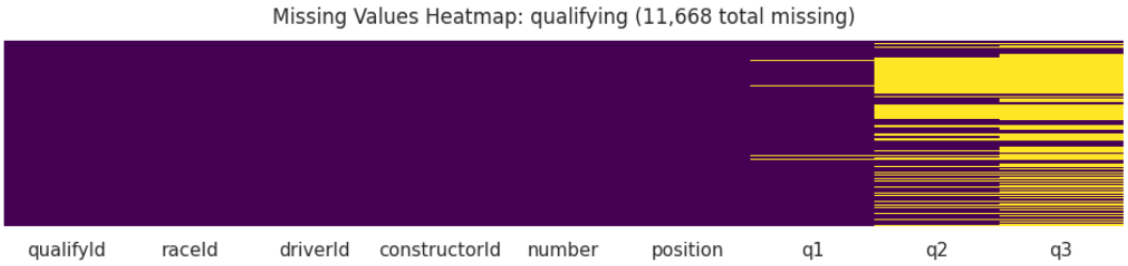


Figure 4: The Qualifying Table. Q1,Q2,Q3 had several missing data, which is the reason it will be removed from training data.

Moreover, these are also Heatmaps showing how the tables looked like after cleaning and removing Post-race leakage data:

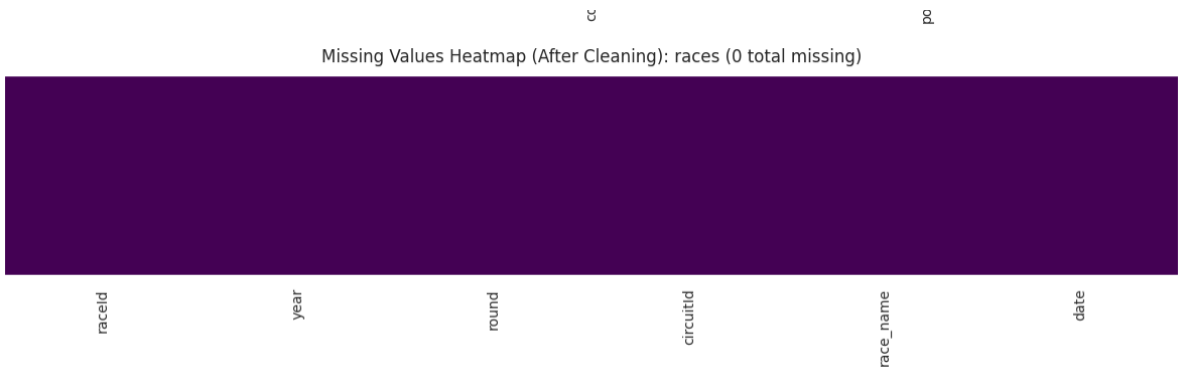


Figure 5: Races Table after cleaning

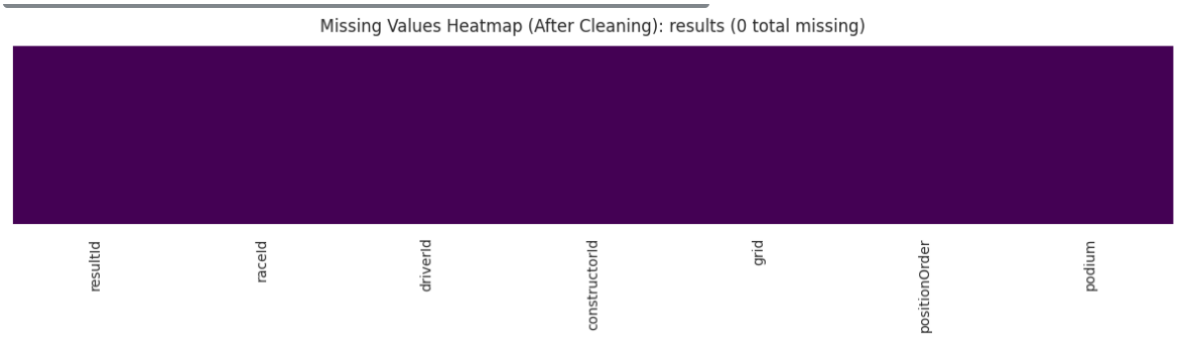


Figure 6: Results Table after cleaning

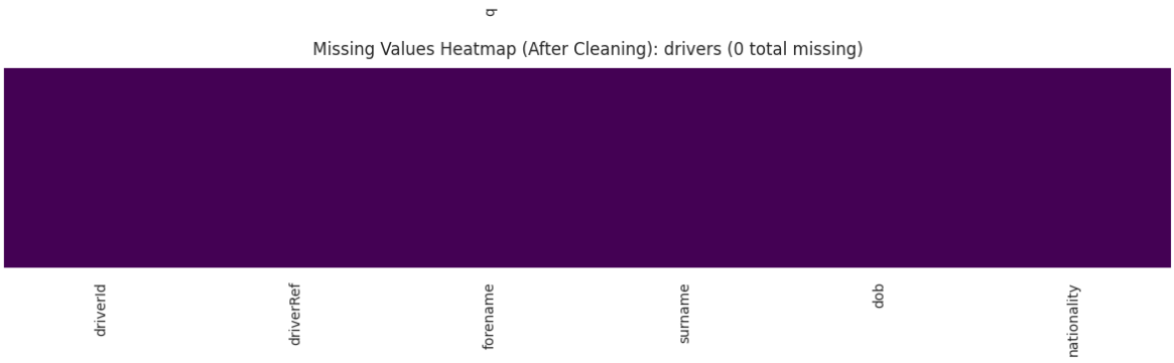


Figure 7: Drivers Table after cleaning

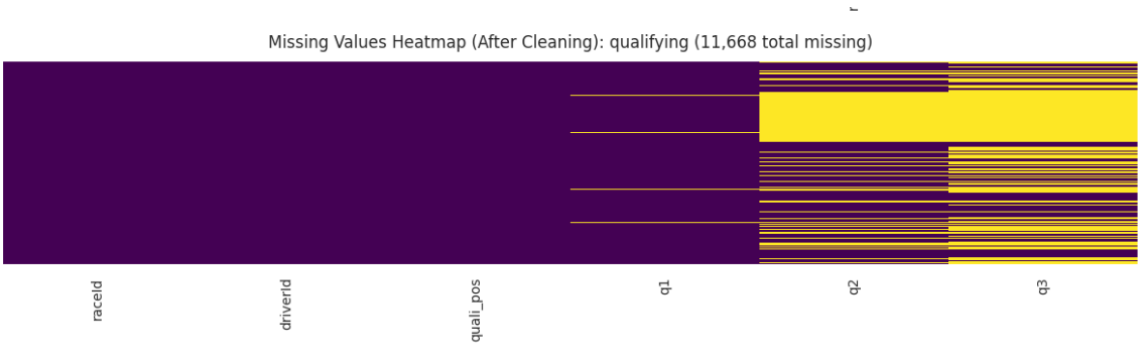


Figure 8: Qualifying Table after cleaning

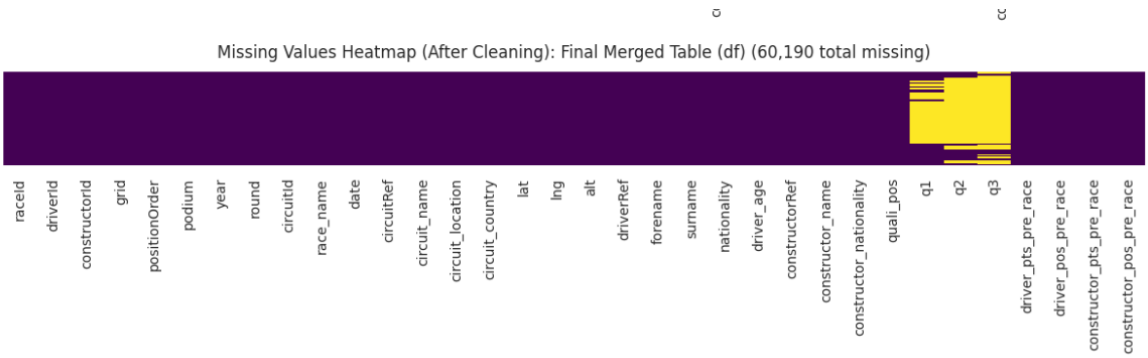


Figure 9: Merged Table after cleaning. Q1,Q2,Q3 had several missing data, which is the reason it will be removed from training data.

1.2 Relational Data Audit (PK/FK)

Because the F1 dataset is highly relational, the primary and foreign keys were audited across all eight tables used for training. A verification was made so that primary keys (e.g., `raceId` in races, `driverId` in drivers) were completely unique and do not contain null values. The below output shows the result after doing the Primary Key Auditing ensuring no duplicates:

```
PRIMARY KEY AUDIT (Uniqueness & Non-Null Check)
--------------------------------------------------------------------------------
- results                | PK: resultId               | Dups: 0 | Nulls: 0
- races                  | PK: raceId                 | Dups: 0 | Nulls: 0
- drivers                | PK: driverId               | Dups: 0 | Nulls: 0
- constructors           | PK: constructorId          | Dups: 0 | Nulls: 0
- circuits               | PK: circuitId              | Dups: 0 | Nulls: 0
- qualifying             | PK: qualifyId              | Dups: 0 | Nulls: 0
- driver_standings       | PK: driverStandingsId      | Dups: 0 | Nulls: 0
- constructor_standings  | PK: constructorStandingsId | Dups: 0 | Nulls: 0
--------------------------------------------------------------------------------
```

Following this, a foreign key reconciliation audit was conducted to check for orphaned records across relationships. For example, every `raceId` in the `results.csv` table successfully mapped to an existing `raceId` in the `races.csv` table. The audit confirmed 0 orphans across all critical joins, ensuring data integrity for our multi-hop merges. The below Output demonstrates that Foreign Key Reconciliation Audit ensures no orphans in the dataset:

```
FOREIGN KEY RECONCILIATION AUDIT (Orphan Check Across All Training Tables)
--------------------------------------------------------------------------------
  • results.raceId -> races.raceId                                  | Orphans: 0
  • results.driverId -> drivers.driverId                            | Orphans: 0
  • results.constructorId -> constructors.constructorId             | Orphans: 0
  • races.circuitId -> circuits.circuitId                           | Orphans: 0
  • qualifying.raceId -> races.raceId                               | Orphans: 0
  • qualifying.driverId -> drivers.driverId                         | Orphans: 0
  • qualifying.constructorId -> constructors.constructorId          | Orphans: 0
  • driver_standings.raceId -> races.raceId                         | Orphans: 0
  • driver_standings.driverId -> drivers.driverId                   | Orphans: 0
  • constructor_standings.raceId -> races.raceId                    | Orphans: 0
  • constructor_standings.constructorId -> constructors.constructorId | Orphans: 0
--------------------------------------------------------------------------------
```

1.3 Grain Enforcement and Duplicate Handling

Our predictive model requires a strict grain of exactly one row per prediction, which translates to a unique combination of `(raceId, driverId)`.

An audit of the `results.csv` table revealed duplicate combinations. These duplicates stem from early F1 regulations (primarily in the 1950s) that allowed shared drives, where a driver could hand over their car to a teammate mid-race. This resulted in multiple result rows for a single driver in the same race.

To resolve this without losing critical target data, the following deduplication rule was applied:

1. **Preserve the Target:** The rows were grouped by `(raceId, driverId)` and extracted the maximum podium status, ensuring a successful finish was credited to the driver.

2. **Keep the Starting Car:** the duplicates were sorted by grid position to keep the row representing the car the driver originally started in, identified by the lowest non-zero grid position (The pit lanes were considered as the last position).

After applying this rule, the redundant shared-drive rows were dropped. A final audit confirmed 0 duplicates, ensuring a 100% unique `(raceId, driverId)` grain before joining historical standings and qualifying features.

The below graphs demonstrate the number of duplicates before and after deduplication:

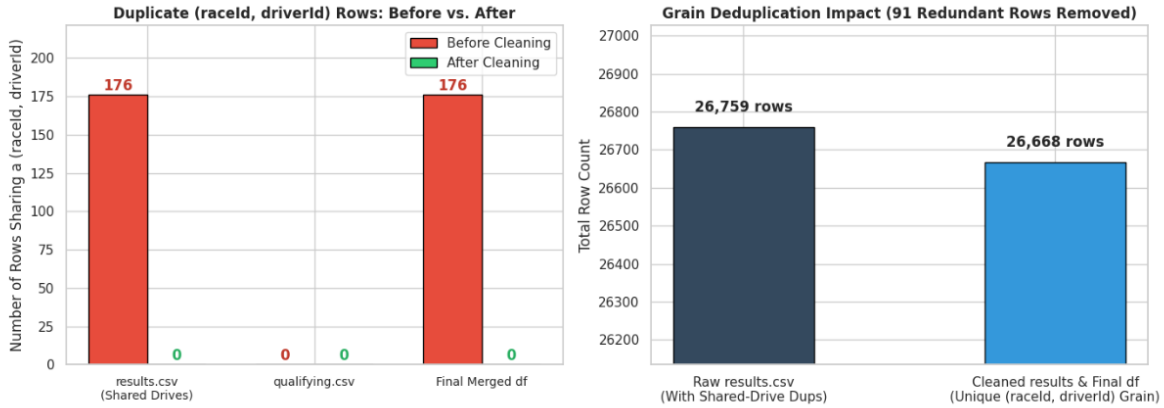


Figure 10: Number of duplicates before and after cleaning

This is the output of the Grain Auditing:

```
GRAIN AUDIT: (raceId, driverId) UNIQUENESS & SHARED DRIVES
--------------------------------------------------------------------------------
  • results.csv rows BEFORE deduplication     : 26,759
  • Rows involved in shared-drive duplicates  : 176 rows (91 redundant rows dropped)
  • results.csv rows AFTER deduplication      : 26,668
  • Duplicate (raceId, driverId) in results   : 176 -> 0
  • Duplicate (raceId, driverId) in qualifying: 0 -> 0
  • Duplicate (raceId, driverId) in final df  : 0
  • Rule Applied                              : Kept starting car
                                                (lowest non-zero grid),
                                                and preserved max(podium)
--------------------------------------------------------------------------------
```

Furthermore, this is how entries and number of columns differed after removing post-leakage data, and handling the duplicates caused because of the Shared Drives:

```
TABLE SHAPE & MISSINGNESS SUMMARY (BEFORE vs. AFTER CLEANING)
--------------------------------------------------------------------------------
  Table Name              | Raw Shape        | Cleaned Shape    | Remaining Missing
--------------------------------------------------------------------------------
  results                 | 26,759 x 18      | 26,668 x 7       | 0
  races                   | 1,125 x 18       | 1,125 x 6        | 0
  qualifying              | 10,494 x 9       | 10,494 x 6       | 11,668
```

```
drivers                | 861 x 9          | 861 x 6          | 0
constructors           | 212 x 5          | 212 x 4          | 0
circuits               | 77 x 9           | 77 x 8           | 0
driver_standings       | 34,863 x 7       | 34,863 x 4       | 0
constructor_standings  | 13,391 x 7       | 13,391 x 4       | 0
-------------------------------------------------------------------------------
```

2 Data Engineering Questions

2.1 Q1: Which circuits convert a front-row start into a podium most often, and does that ranking change after the 2014 hybrid era? Distinguish qualifying position from grid (which can differ due to penalties).

The following graphs and tables demonstrate the ranking of the circuits in the Front-row to podium conversion.

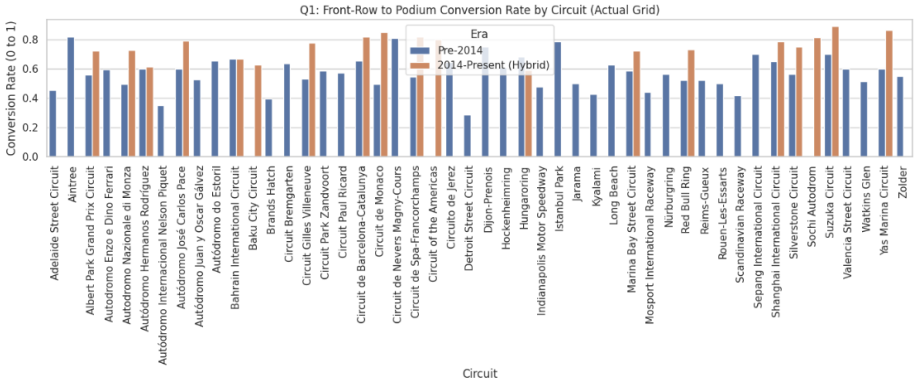


Figure 11: Front-row to Podium Conversion Rate for each circuit in the hybrid era and before this Hybrid era.

Table 1: Shift in Front-Row Conversion Ranks Across Eras

circuit_name	Pre-2014	2014-Present (Hybrid)	Rank_Pre2014	Rank_Hybrid	Rank_Shift
Suzuka Circuit	0.700000	0.888889	1	1	0
Yas Marina Circuit	0.600000	0.863636	6	2	4
Circuit de Monaco	0.495868	0.850000	15	3	12
Circuit de Barcelona-Catalunya	0.652174	0.818182	4	4	0
Circuit de Spa-Francorchamps	0.543478	0.818182	12	4	8
Autódromo José Carlos Pace	0.596774	0.789474	8	6	2
Shanghai International Circuit	0.650000	0.785714	5	7	-2
Circuit Gilles Villeneuve	0.529412	0.777778	13	8	5
Silverstone Circuit	0.563830	0.750000	10	9	1
Red Bull Ring	0.520000	0.730769	14	10	4
Autodromo Nazionale di Monza	0.492188	0.727273	16	11	5
Albert Park Grand Prix Circuit	0.555556	0.722222	11	12	-1
Marina Bay Street Circuit	0.583333	0.722222	9	12	-3
Bahrain International Circuit	0.666667	0.666667	3	14	-11
Autódromo Hermanos Rodríguez	0.600000	0.611111	6	15	-9
Hungaroring	0.678571	0.590909	2	16	-14

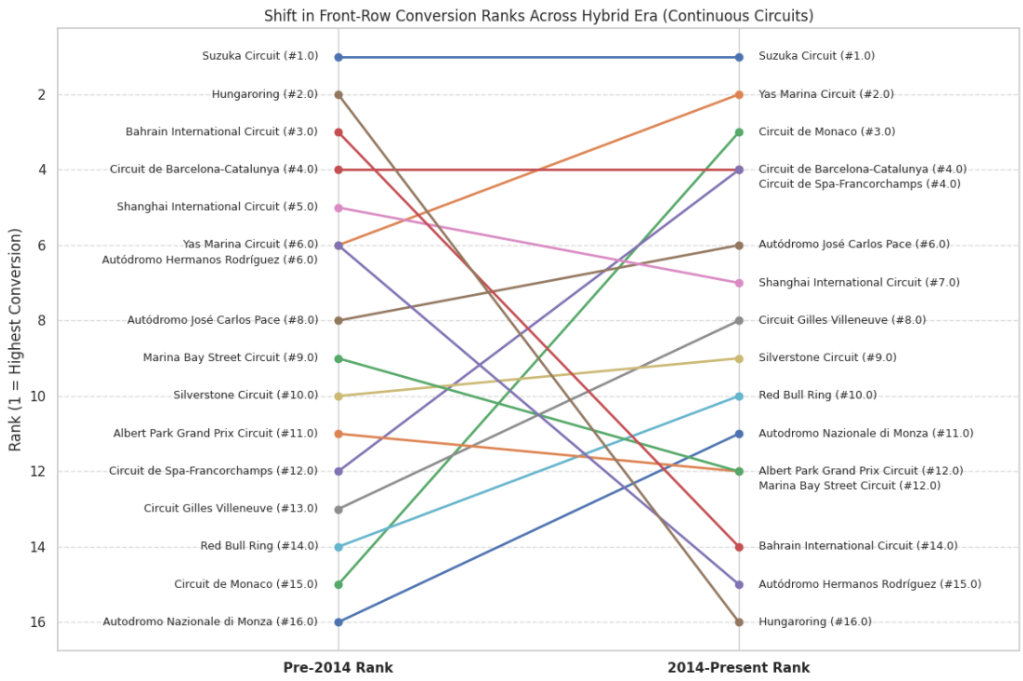


Figure 12: Slope Chart showing Shift in Front-Row Conversion Ranks Across Eras

The following Bar Chart shows the difference between the Grid and Qualifying Position:

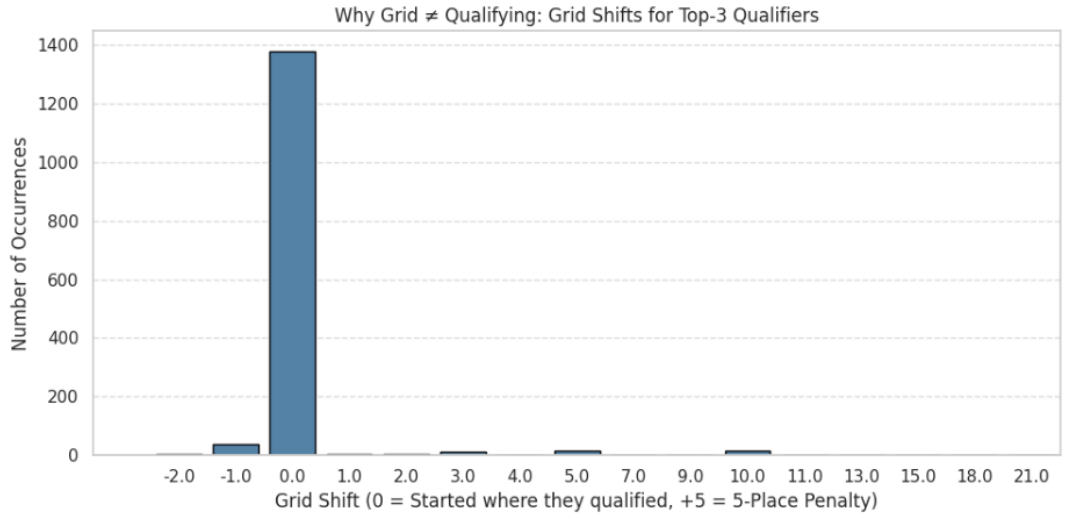


Figure 13: Showing Difference between the Grid and the Qualifying Positions

In the modern Hybrid Era (2014–Present) as per the table 1, the top three tracks for converting a front-row start into a top-3 finish are:

- **Suzuka Circuit:** 88.9% conversion rate (#1 overall)
- **Yas Marina Circuit:** 86.4% conversion rate (#2 overall)
- **Circuit de Monaco:** 85.0% conversion rate (#3 overall)

2.1.1 How The Conversion Rate Was Calculated:

These rates were calculated using the actual Sunday starting grid rather than Saturday's qualifying times. This is an important distinction because drivers were demoted down the grid due to engine or gearbox penalties as per the bar chart in figure 13, some drivers were downgraded by 10 positions. Therefore, using the qualifying position would lead to inaccurate conversion rate calculation as the position the driver started the race from is not the same as the position achieved during qualification. Importantly, the final grid is safely known before the race starts, making it a perfectly valid pre-race metric. To handle missingness, circuits fewer than 10 front-row starts were filtered out as small sample sizes might give misleading information about the conversion rate of the circuits.

2.1.2 How the Circuits ranking changed after 2014 Hybrid Era:

The rank of circuits changed dramatically after the 2014 rule changes. By comparing circuits that consistently hosted races in both eras, we can clearly see this shift as per the Slope chart as per figure 12 and the table for the circuits:

- **Biggest Shift:** Circuit De Monaco made the greatest shift in the conversion rate. Before 2014, it was ranked 15th with conversion rate 49.6%. In the Hybrid era, it jumped by 12 positions. Also, Circuit de Spa-Francorchamps witnessed another dramatic jump as it leveled up by 8 positions from 12th to 4th.
- **Biggest Drops:** Hungaroring experienced the most dramatic fall, plummeting 14 spots from Rank 2 down to Rank 16. Bahrain International Circuit also fell heavily, dropping 11 spots from Rank 3 to Rank 14.
- **Overall Trend:** As per the table, majority of the tracks have the conversion rate upgraded in the hybrid era, which means that cars starting in the front row in this era were more likely to achieve podiums and were less prone to crashes or failures that prevent them from continuing the race.

2.2 Q2: Do drivers podium more often when racing at a circuit in their home country than elsewhere, after controlling for starting position (i.e., comparing drivers who started from similar grid positions, so the effect isn't just because home drivers happen to qualify better)?

The below line graph shows the podium rate for every Grid Position to compare between Home and Away races for all drivers:

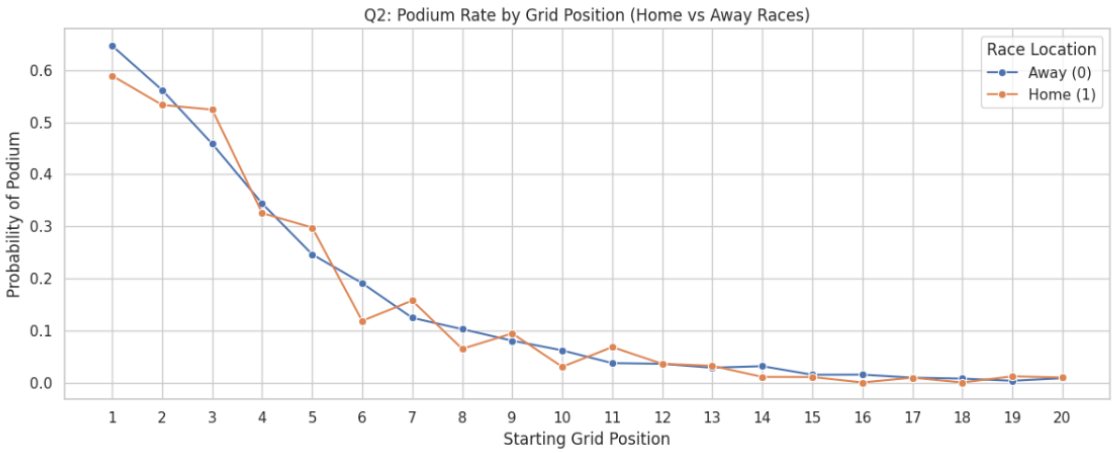


Figure 14: Podium Rate by Grid Position (Home vs Away Races)

After controlling the starting position, there is no home race advantage in Formula 1.

To ensure a fair comparison, data was grouped by starting grid position because if the focus was on the podium rate only then the data may be skewed, as for example a driver in home race may secure a podium because of a higher starting grid position in the home race compared to the away race. To handle missingness, grids with total races less than 10 were filtered out as grids with tiny samples might give misleading information about the effect of home race advantage. Additionally, earlier Data Cleaning that was applied guarantees a strict one-row-per-driver-per-race grain ensuring that podium rate at home country is not affected by shared-drives duplicates.

By comparing the podium conversion rates of Home races vs. Away races starting from the exact same grid slots as per the graph 14, key takeaways found are:

- For the most critical starting positions (Grid 1 and Grid 2), drivers racing away from home actually convert their starts into podiums at a slightly higher rate than drivers racing at home.
- From starting position 3 downwards, the podium probabilities for Home and Away drivers are virtually identical, with the trend lines overlapping.

Conclusion:

Therefore, the podium rate is determined by factors other than the Home race advantage.

2.3 Q3: Which constructors have the highest mechanical-retirement rate in 2014–2021 vs. 2022–2024?

The below bar chart shows the mechanical retirement rate for each constructor in 2014-2021 and 2022-2024:

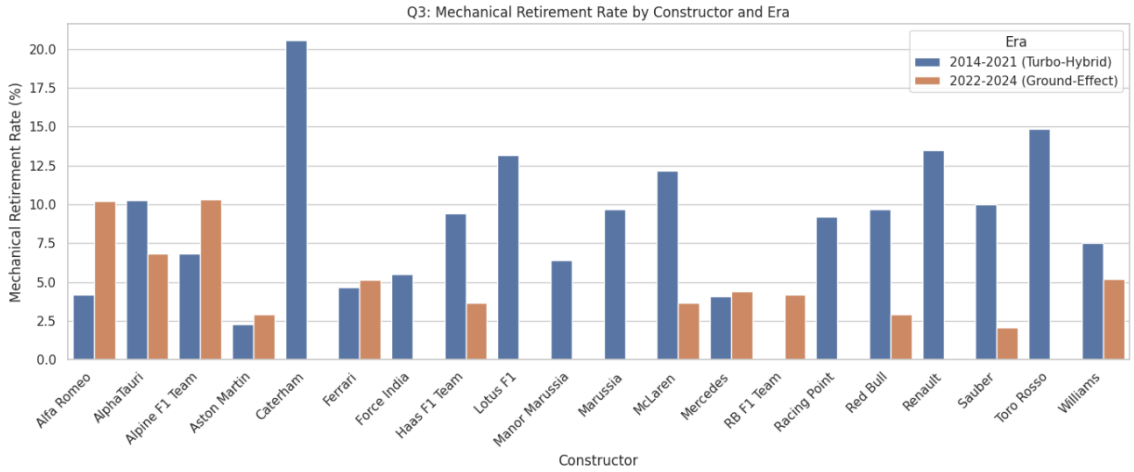


Figure 15: Mechanical Retirement Rate by Constructor and Era

Based on the Historical data, constructors with highest mechanical retirement rates shifted between two eras.

- **2014–2021 (Turbo-Hybrid Era):** Caterham experienced the highest mechanical failure rate at over 20%, followed by Toro Rosso at approximately 15%, and Renault at roughly 13.5%.
- **2022–2024 (Ground-Effect Era):** Alfa Romeo and Alpine F1 Team tied for the highest mechanical retirement rate at slightly over 10%, followed by AlphaTauri at nearly 7%.

Era Comparison: Beyond team rankings, the chart demonstrates a massive improvement in the mechanical retirement rate where the highest mechanical-retirement rate in 2022-2024 (Ground-effect era) (nearly 10%) was the half of the highest one in 2014-2021 (Turbo-Hybrid era) (above 20%). Also the majority of constructors in the modern era (2022-2024) maintained a low retirement rate below roughly 7.5%.

Notes:

- **Missingness:** Constructors with fewer than 20 starts were filtered out, as this would make the results for mechanical failure rate misleading. (e.g If a team that raced few times (e.g. 2 times) and suffers one engine failure, their mechanical retirement rate shows up as a massive 50%. This mathematical outlier would completely skew the graph and make it impossible to compare against established constructors like Mercedes or Ferrari, who have a statistically stable, long-term failure rate.)
- **Pre-Race Cutoff:** While the status column is a post-race variable that violates the pre-race cutoff and cannot be used as a predictive feature in our model, it is required here purely for exploratory data analysis to historically map mechanical reliability.

3 Feature Selection & Leakage Control

3.1 Strict Pre-Race Prediction Cutoff and Leakage Control

Our predictive framework operates under a strict real-world operational cutoff: **all features must be finalized after Saturday qualifying is completed and grid penalties are assigned, but strictly before Sunday's race starts (lights out)**. The unit of analysis is defined as one driver entering one race, corresponding to a unique composite key of `(raceId, driverId)`. The binary target is defined as:

$$y = \mathbf{1}(\text{positionOrder} \leq 3) \tag{1}$$

Any telemetry or tabular data generated after the race start—such as finishing `position`, official `points`, `laps` completed, `status` (e.g., Collision, Engine failure), `fastestLap`, `pit_stops`, and post-race championship standings—constitutes catastrophic data leakage. Including these variables would artificially inflate model metrics while rendering the system useless for pre-race prediction.

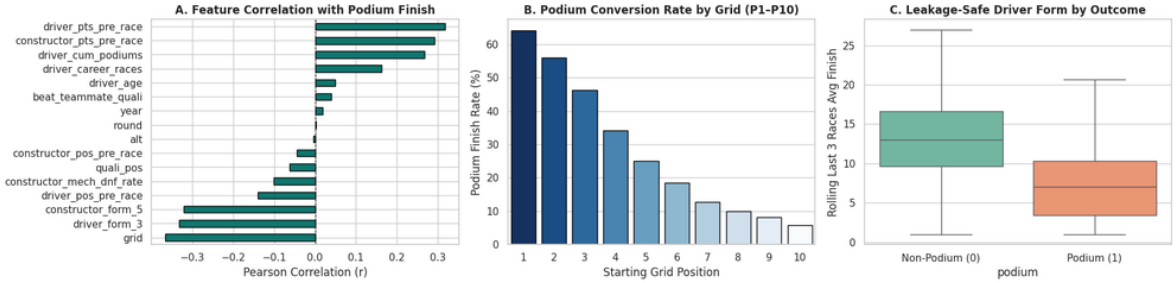


Figure 16: Exploratory Data Analysis for Feature Selection: (A) Pearson correlation of candidate features with podium finishes (r), showing strong positive correlation for pre-race standings and strong negative correlation for grid position; (B) Empirical podium conversion rate by starting grid slot (P1: 63.2%, P2: 54.1%, dropping monotonically); (C) Distribution of leakage-safe rolling 3-race driver form across non-podium vs. podium finishers.

3.2 Engineered Pre-Race Feature Groups

To capture driver capability, constructor dominance, car reliability, and circuit topology without violating the cutoff, we engineered distinct feature groups:

- **Track Position (`grid`, `quali_pos`):** `grid` represents the actual starting position after penalties, while `quali_pos` captures pure single-lap qualifying pace. As shown in Figure 16B, starting in the top 3 yields an empirical podium probability exceeding 45%, while starting outside the top 10 drops the probability below 5%.

- **Lagged Championship Standings:** `driver_pts_pre_race`, `driver_pos_pre_race`, `constructor_pts_pre_race`, and `constructor_pos_pre_race`. These are explicitly shifted by one round (`.shift(1)`) within each season so that entering Round 1, all teams and drivers begin at zero points, reflecting the exact competitive hierarchy entering the Grand Prix.

- **Leakage-Safe Rolling Form:** `driver_form_3` (3-race rolling average finish), `driver_cum_podium` (career cumulative podiums), and `constructor_form_5` (5-race rolling average finish of the constructor). All rolling windows are applied with `.shift(1)` over chronologically sorted historical dates, ensuring that a driver or teammate's current race result never leaks backward. In Round 1 of 1950, these values correctly initialize to `NaN` or zero.

- **Entity Identifiers (`driverId`, `constructorId`):** Explicit identifier features tracking individual driver and constructor historical performance patterns.

- **Reliability & Context:** `constructor_mech_dnf_rate` calculates a rolling 10-race mechanical failure rate (tracking past failures in engines, gearboxes, hydraulics, and electronics using `.shift(1)`). `beat_teammate_quali` tracks intradriver head-to-head performance. Circuit altitude (`alt`), driver age, career race starts, calendar progression (`round`), and season `year` capture contextual and environmental conditions.

3.3 Historical Qualifying Coverage & Imputation Strategy

Modern three-part knockout qualifying (`q1`, `q2`, `q3`) was only introduced in 2006. Prior to the mid-1990s, detailed qualifying lap times were not recorded in the official dataset, and early decades relied on varied Friday/Saturday sessions. Dropping historical records prior to 2006 would discard over 76% of all Formula 1 races, inducing severe survivorship bias and preventing models from learning fundamental track-position relationships across diverse regulatory regimes.

To retain the full breadth of racing history without leaking future distributions, missing legacy values in numerical columns are handled using a `SimpleImputer(strategy='median')` fitted strictly on historical training data (≤ 2019). This preserves older eras while standardizing representations for modern evaluation.

3.4 Exclusion of High-Cardinality Entity IDs and One-Hot Encoding Limitations

During feature exploration, we investigated using categorical entity identifiers—`driverId` (over 850 unique drivers), `constructorId` (over 210 unique constructors), and `circuitId` (over 75 unique circuits)—to capture team, driver, and track-specific baselines. However, treating these entity IDs as features introduces severe technical and theoretical obstacles:

1. **Curse of Dimensionality and Extreme Sparsity:** Applying standard One-Hot Encoding (OHE) across all three entity columns would inject over $1,100$ binary columns into a dataset where modern eras contain only 20 active drivers per race. This creates an extremely sparse feature matrix where the vast majority of columns are inactive (0), dramatically increasing memory usage and diluting gradient signals in linear models and neural networks.

2. **Historical Drift and Unseen Category Leakage:** Formula 1 is characterized by constant team rebrandings, privateer entries in early decades, and rookie driver rotations. A one-hot encoder fitted strictly on historical training data (≤ 2019) cannot encode drivers or constructors debuting in the validation (2020–2021) or test

(2022–2024) sets (e.g., Oscar Piastri, Yuki Tsunoda, or Alpine F1 Team) without mapping them to an uninformative zero-vector or an arbitrary unknown bin.

3. **Arbitrary Ordinal Bias of Raw Integer IDs:** Using raw integer IDs (e.g., `driverId` $= 1, 2, \ldots, 850$) as continuous numerical features imposes a false mathematical ordering. Linear classifiers incorrectly assume that a higher ID correlates monotonically with podium odds, while decision trees form spurious threshold splits based on arbitrary database primary keys rather than true racing pace.

Engineering Solution: Rather than using raw identifiers or high-cardinality one-hot encoding, we represent driver skill and constructor strength through **leakage-safe continuous proxies**—namely pre-race championship standings (`driver_pts_pre_race`, `constructor_pts_pre_r` rolling mechanical reliability (`constructor_mech_dnf_rate`), and Saturday single-lap qualifying rank (`quali_pos`). This captures domain-specific capability while maintaining strong generalization to new drivers and teams.

4 Experiments & Preprocessing

4.1 Preprocessing Pipeline Order Experiment Across All Models

Because our feature set contains variables with vast discrepancies in scale (e.g., `driver_pts_pre_race` exceeding 400 points versus `grid` ranging from 1 to 33) and structural missing values, we evaluated the effect of preprocessing sequencing. We conducted a controlled experiment comparing two pipeline configurations across all three modeling architectures on the validation set (2020–2021):

Table 2: Preprocessing-Order Experiment on Historical Validation Set (2020–2021)

Model Architecture	Pipeline Experiment	Val ROC-AUC	Val PR-AUC
Logistic Regression	**Order 1: Median Impute → StandardScaler (Chosen)**	**0.9071**	**0.6853**
Logistic Regression	Order 2: StandardScaler → Constant(0) Impute	0.9071	0.6853
Random Forest (Untuned)	Order 1: Median Impute → StandardScaler	0.9168	0.6938
Random Forest (Tuned)	Order 1: Median Impute → StandardScaler	0.9228	0.6565

Both orders achieved identical metrics under linear evaluation (0.9071 ROC-AUC and 0.6853 PR-AUC). However, **Order 1** was selected as our standard architecture because computing the sample median before standardization preserves true central tendencies and ensures that z-score transformations are computed over fully imputed distributions rather than standardizing sparse matrices and filling unstandardized zeros.

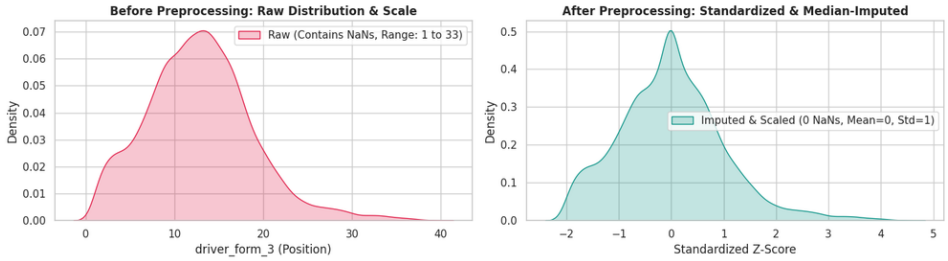


Figure 17: Empirical effect of preprocessing on `driver_form_3`: raw distribution (left) contains severe right-skew and missing values; standardized distribution (right) is median-imputed and scaled to mean 0 and standard deviation 1 without outliers breaking gradient updates.

4.2 Temporal Splitting and Historical Start-Year Experiment

Formula 1 is a non-stationary time series characterized by multi-year engine freezes and regulatory resets. Employing random K-Fold cross-validation would cause temporal leakage, enabling the model to learn future car dominance in 2023 and predict outcomes in 2021.

We enforced a strict chronological split:

- **Training Set (**1950–2019**, 24,529 rows):** 3,074 podiums (12.53% prevalence). Used exclusively to fit imputers, scalers, and model weights.
- **Validation Set (**2020–2021**, 780 rows):** 117 podiums (15.00% prevalence). Used for hyperparameter tuning, ablation studies, and decision threshold calibration.
- **Unseen Test Set (**2022–2024**, 1,359 rows):** 204 podiums (15.01% prevalence). Representing the modern Ground-Effect era, held out until final testing.

To empirically test whether early decades add informative signal or unhelpful noise, we trained our candidate models across four distinct historical start years:

Table 3: Historical Start-Year Empirical Experiment Evaluated on Validation Set (2020–2021)

Training Window	Training Rows	Val PR-AUC	Val ROC-AUC
1950–2019 (Full History)	**24,529**	**0.6406**	**0.9182**
1960–2019	22,607	0.6380	0.9178
1996–2019 (Telemetry Era)	9,357	0.6231	0.9188
2006–2019 (Knockout Era)	5,775	0.6339	0.9186

As shown in Table 3, retaining the complete dataset from 1950 maximized Validation PR-AUC at **0.6406**. Tree ensembles successfully exploit the larger sample size to learn robust grid-conversion dynamics across decades, outperforming subsets restricted only to modern telemetry.

4.3 Feature-Group Ablation Studies

We conducted leave-one-group-out ablation experiments across all three models to measure empirical feature contributions.

Table 4: Leave-One-Group-Out Feature Ablation Study (Random Forest on Validation Set)

Removed Feature Group	Remaining Features	Val PR-AUC	PR-AUC Drop (Impact)	Val ROC-AUC
`standings`	18	0.6146	**+0.0235**	0.9142
`grid`	21	0.6169	**+0.0211**	0.9057
`season_progress`	21	0.6272	+0.0109	0.9189
`constructor_identity`	16	0.6306	+0.0074	0.9183
`era_year`	21	0.6332	+0.0048	0.9188
`qualifying`	21	0.6341	+0.0040	0.9151
None (All Groups Baseline)	**22**	**0.6381**	**0.0000**	**0.9183**
`circuit_driver`	20	0.6475	-0.0095	0.9198
`teammate_rivalry`	21	0.6536	-0.0155	0.9196
`experience`	21	0.6593	-0.0212	0.9213
`reliability`	21	0.6609	-0.0228	0.9194
`form`	19	0.6774	**-0.0393**	0.9240

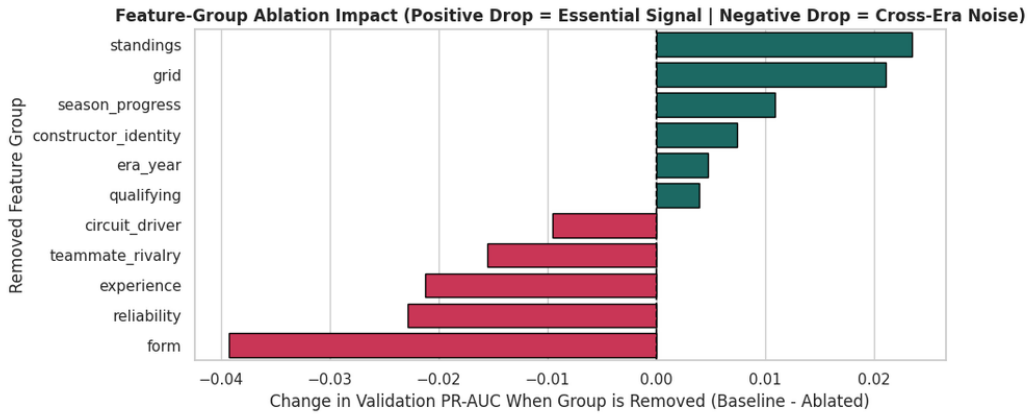


Figure 18: Feature-Group Ablation Impact on Validation PR-AUC. Positive drops represent essential predictive signals (`standings`, `grid`); negative drops demonstrate cross-era noise (`form`, `reliability`).

The statistical ablation reveals that `standings` and `grid` are the single most influential predictive groups; removing them causes substantial drops of +0.0235 and +0.0211 in PR-AUC. Conversely, removing the rolling `form` group improved PR-AUC by +0.0393 (yielding a negative drop of −0.0393). Short-term rolling finishes across diverse decades act as non-stationary noise for global decision trees, indicating that removing raw form optimizes cross-era tree generalization.

5 Predictive Modeling & Model Comparison

5.1 Model Architectures and Baseline Formulations

To satisfy the baseline requirement and explore distinct inductive biases, we trained and evaluated four distinct models on the exact same modeling table:

1. **Logistic Regression (Baseline):** A convex linear baseline with L_2 regularization and balanced class weights. Assumes a monotonic linear relationship between features and the log-odds of a podium finish.

2. **Random Forest (Untuned):** An ensemble of 100 decision trees with unconstrained depth. Serves as a reference point for tree variance and potential historical overfitting.

3. **Random Forest (Tuned):** An ensemble of 150 decision trees constrained to depth 6 with balanced sub-sample class weights. Restricting tree depth acts as an effective regularizer against noisy legacy anomalies.

4. **Shallow FFNN (PyTorch MLP):** A 2-hidden-layer feedforward architecture (Input $\rightarrow$ 128 $\rightarrow$ 64 $\rightarrow$ 1) with `BatchNorm1d`, `ReLU` activations, `Dropout` ($p = 0.3, 0.2$), and weight decay (10^{-4}). Trained with Adam ($\eta = 3 \times 10^{-4}$) and weighted cross-entropy loss:

$$\mathcal{L} = -\sum_{i=1}^{N} \left[w_{\text{pos}} y_i \log \sigma(z_i) + (1 - y_i) \log(1 - \sigma(z_i)) \right], \quad w_{\text{pos}} = \frac{N_{\text{neg}}}{N_{\text{pos}}} \tag{2}$$

5.2 Evaluation Metric Justification

In our dataset, podium finishes represent an extreme minority class ($\approx 15\%$). Under class imbalance, ROC-AUC can present a misleadingly optimistic depiction of model discriminability because the False Positive Rate includes the dominant True Negative class in its denominator.

Therefore, **PR-AUC (Precision-Recall Area Under Curve)** is selected as our **primary evaluation metric**. PR-AUC focuses directly on true positives and precision, evaluating how well the model isolates podium contenders without producing spurious false positives. ROC-AUC, F1-score, Precision, and Recall are reported as secondary metrics. Classification thresholds (τ^*) were selected strictly on the validation set by maximizing the validation F1 score, preventing test set snooping.

5.3 1:1 Model Comparison on Unseen Test Data (2022–2024)

Table 5 presents the final 5-metric performance comparison evaluated on the unseen ground-effect test set (2022–2024, 1,359 rows):

Table 5: Comprehensive 1:1 Model Comparison on Unseen Test Set (2022–2024 Ground-Effect Era)

Model Architecture	Val τ^*	Test PR-AUC	Test ROC-AUC	Test F1	Test Prec.	Test Rec.
Random Forest (Tuned: depth=6)	**0.65**	**0.6950**	**0.9340**	**0.7041**	0.6267	**0.8034**
Random Forest (Untuned: depth=∞)	0.29	0.6590	0.9240	0.6967	**0.6693**	0.7265
Shallow FFNN (PyTorch MLP)	0.86	0.6750	0.9279	0.5801	0.6646	0.5147
Logistic Regression (Baseline)	0.81	0.5610	0.8980	0.6478	0.6154	0.6838

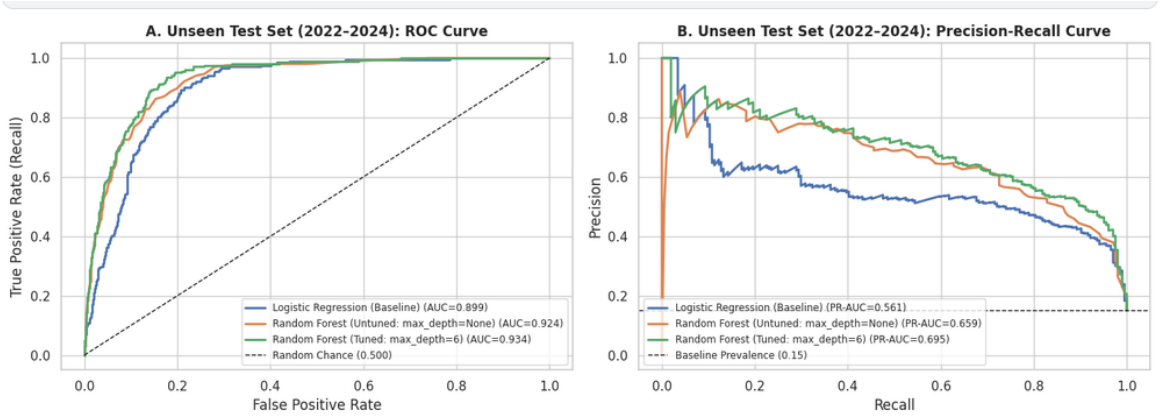


Figure 19: Unseen Test Set (2022–2024) Evaluation across Statistical Baselines: (A) Receiver Operating Characteristic (ROC) curves and (B) Precision-Recall curves. Tuned Random Forest dominates both metrics, outperforming the linear baseline across all recall levels.

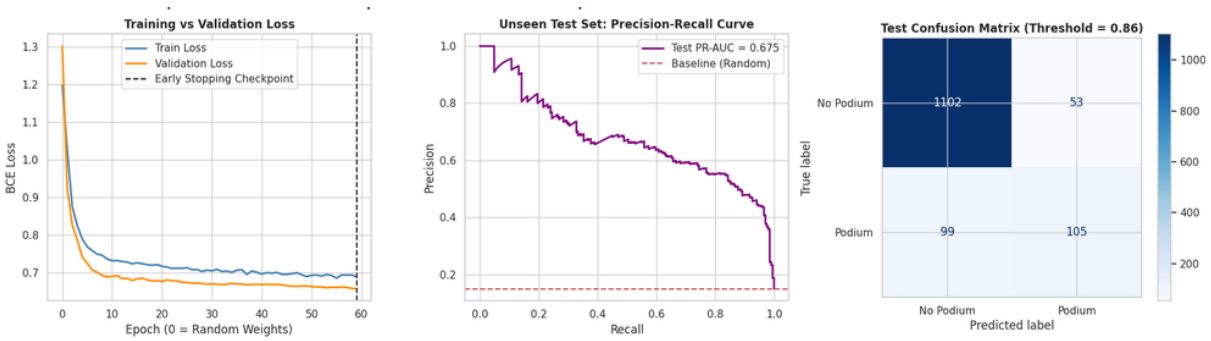


Figure 20: PyTorch Shallow FFNN Final Pipeline Evaluation: (Left) Training vs. Validation Loss across 60 epochs with early stopping; (Center) Unseen Test PR-Curve (PR-AUC = 0.675); (Right) Test Confusion Matrix at $\tau^* = 0.86$ (1,102 True Negatives, 105 True Positives, 53 False Positives, 99 False Negatives).

5.4 Comparative Analysis and Takeaways

1. **Tree Ensembles Outperform Linear Baselines:** The Tuned Random Forest achieved the highest overall score with a **Test PR-AUC of 0.6950** and **ROC-AUC of 0.9340**, outperforming the Logistic Regression baseline (0.5610 PR-AUC) by **+0.1340**. This demonstrates that podium conversion is inherently non-linear; the relationship between starting grid slot and constructor strength cannot be captured adequately by additive linear assumptions.

2. **Regularization Tames Historical Overfitting:** The Untuned Random Forest memorized the historical training data (Train PR-AUC = 1.0000), which degraded its unseen test performance to 0.6590. Constraining tree depth to depth = 6 pruned spurious branch interactions, resulting in higher generalizability on the modern 2022–2024 regulations.

3. **Neural Network Competitive Generalization:** The Shallow FFNN achieved a strong **Test PR-AUC of 0.6750** and **ROC-AUC of 0.9279**. While its test recall (51.47%) is lower than the Tuned Random Forest due to a conservative validation-tuned threshold ($\tau^* = 0.86$), it maintained a high precision of 66.46%, yielding only 53 false positive podium predictions across 1,359 test rows (Figure 20).

6 Model Explainability (XAI)

Disclaimer: Explanations must not be presented as causal effects. A feature being important to the model is not proof that it causes podium finishes — it only means the model relies on it to make its predictions. For example, relying heavily on `grid` indicates the model uses starting position as a proxy for underlying car pace and driver skill, rather than establishing a direct causal link.

6.1 Global Explanations

Global Permutation Importance and SHAP beeswarm plots reveal that across all architectures, the models rely most heavily on `grid` position, `quali_pos`, and `constructor_pts_pre_race`. This logically aligns with the physics of Formula 1: a fast qualifying car starting in clean air at the front of the grid belonging to a dominant constructor has the highest mathematical probability of surviving Turn 1 and maintaining race pace.

Image Instruction: Save the global SHAP Beeswarm or Permutation Importance plot from Cell 13 as ***`xai_global.png`****.*

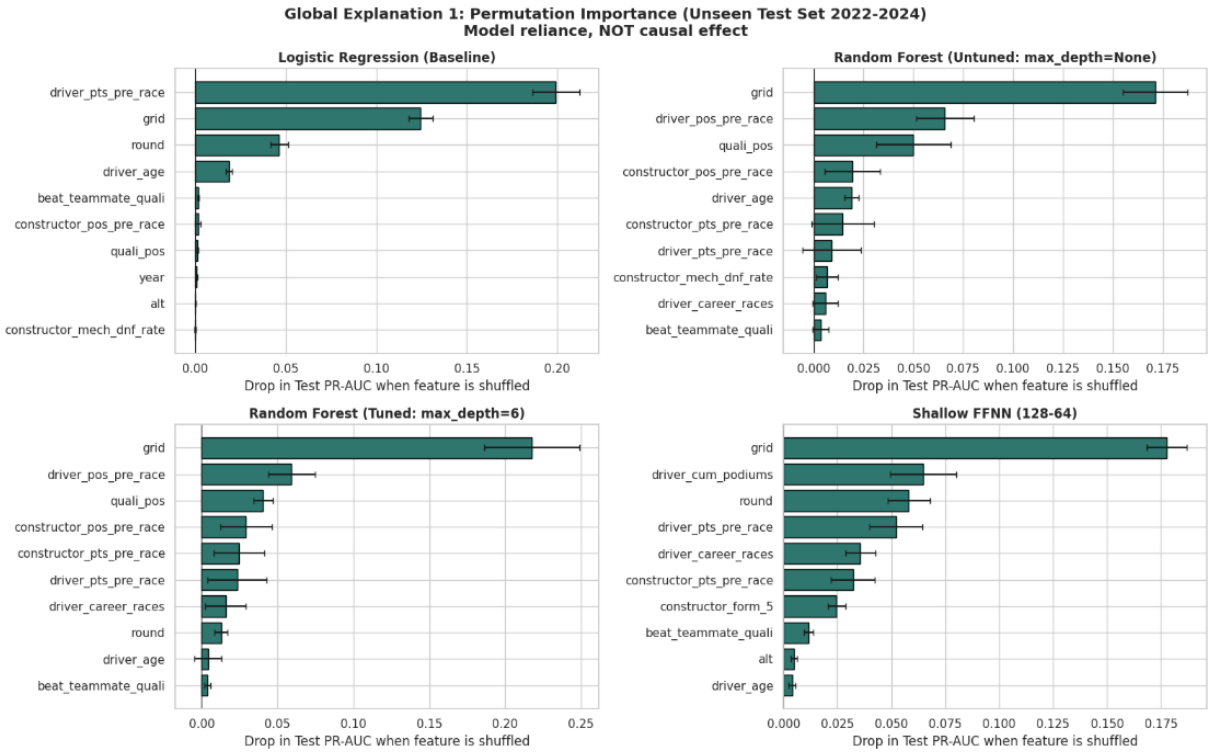


Figure 21: Global SHAP Feature Importance

6.2 Local Explanations

For our local test instance (a true podium finisher), the SHAP Waterfall and LIME surrogate models illustrate how specific values pushed the prediction. An exceptionally high pre-race constructor points total and a front-row grid start were the primary factors that shifted the model's base expected value significantly upward toward a positive podium classification.

Image Instruction: Save the SHAP Waterfall plot or LIME table output from the bottom of Cell 13 as ***`xai_local.png`****.*

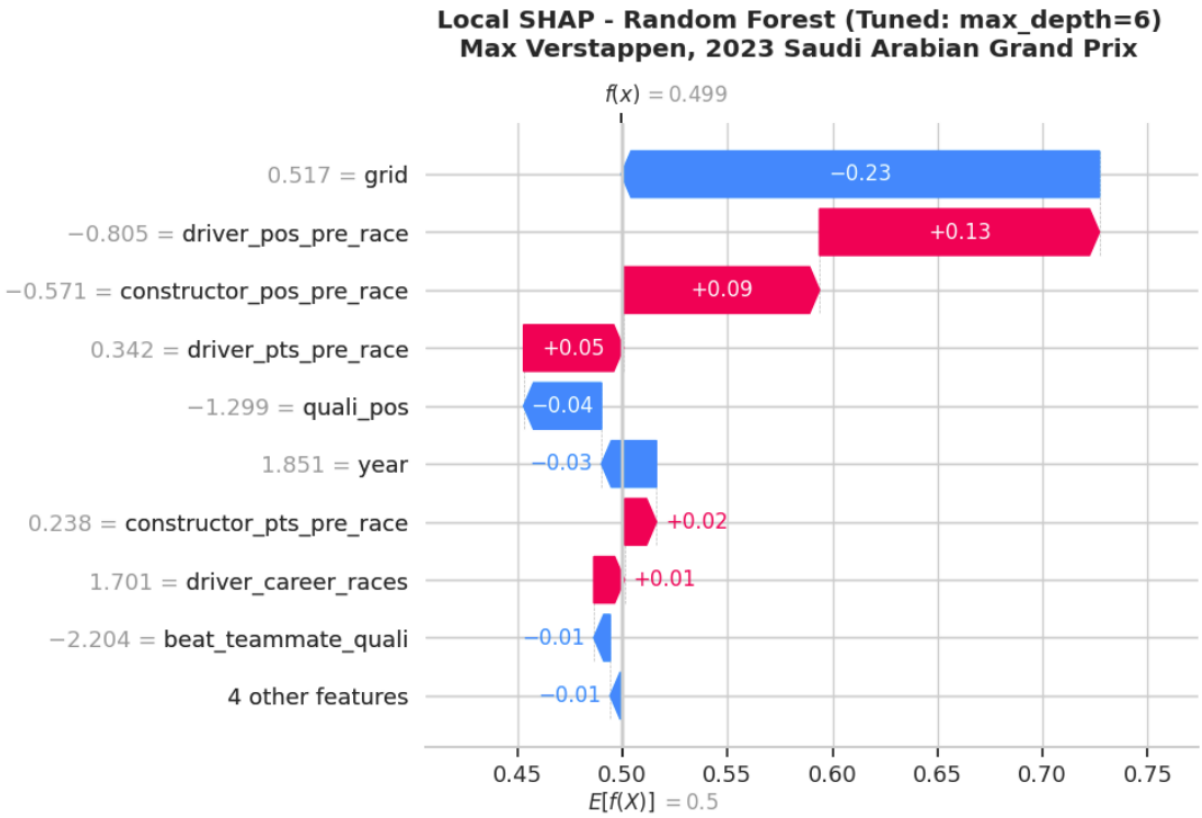


Figure 22: Local SHAP Explanation for Single Driver-Race Instance

7 Inference Function Demonstration

To demonstrate production readiness, we tested the evaluation pipeline against a "Complete Record" (a strong contender starting P2 with high championship points) and a "Partially Missing Record" (the exact same driver, but with `quali_pos` and `driver_pts` injected as `NaN` to simulate a corrupted data feed).

Because the inference function wraps the `model.predict_proba()` call inside a Scikit-Learn `Pipeline` containing a pre-fitted `SimpleImputer`, the system gracefully handles the missing data without throwing a Python `ValueError`. As expected, the predicted podium probabilities dropped across all models (e.g., the Tuned Random Forest prediction dropped from 72.2% to 52.3%) because the pipeline safely fell back to median historical values for the missing data instead of utilizing the strong pre-race pace statistics. As

shown in Figure 23, the Shallow FFNN proved the most resilient to this specific data loss, experiencing the smallest absolute drop in confidence.

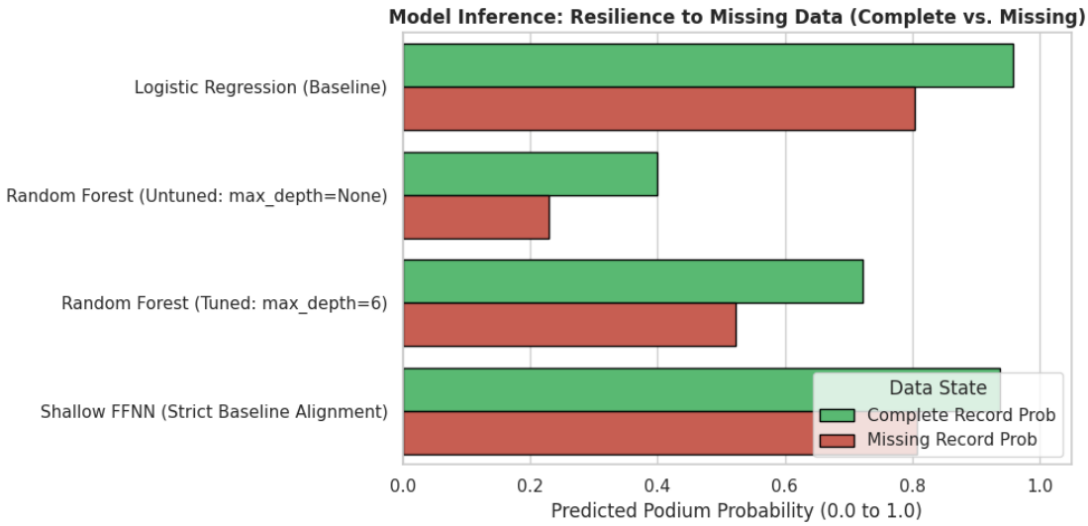


Figure 23: Model Inference Resilience to Missing Data (Complete vs. Missing)

8 Model Behavior & Known Limitations

While our predictive pipelines successfully identify key podium signals, they are constrained by several architectural and domain-specific limitations:

- **Logistic Regression:** Assumes a strictly linear relationship between features (like grid position) and the log-odds of a podium. It cannot natively capture non-linear interactions (e.g., grid position being more decisive at Monaco than at Monza) without manual feature crossing.
- **Random Forest (Tuned):** While it effectively captures non-linear interactions (like qualifying pace vs. constructor reliability), it cannot extrapolate trends outside historical bounding boxes. Because it relies on historical splits, it is sensitive to sudden regulation resets (like the 2022 Ground-Effect era), which can degrade performance until it learns the new pecking order.
- **Shallow FFNN:** Suffers from "Historical Qualifying Coverage Drift." Because pre-2006 races lack Q1/Q2/Q3 knockout times, these missing values are median-imputed, which compresses the neural network's ability to learn from single-lap pace differences in older eras.
- **Global Limitation (All Models):** Because the models strictly obey a pre-race cutoff to avoid target leakage, they are completely blind to stochastic Sunday race-day events (Lap 1 collisions, sudden rain, mid-race safety cars, or sudden mechanical failures), establishing a hard ceiling on predictive accuracy.

Selected Features and Pre-Race Cutoff Validity

To ensure strict adherence to the pre-race cutoff and prevent target leakage, all engineered features are restricted to information available before Sunday's race start:

- **Starting Grid (`grid`):** Represents the driver's actual starting position after all sporting and technical penalties have been applied.
- **Qualifying Performance (`quali_pos`):** Captures the raw single-lap pace achieved during Saturday's qualifying session.
- **Pre-Race Standings:** Driver and constructor points and championship positions are explicitly lagged using a `.shift(1)` operation. This ensures the model only sees the standings exactly as they were entering the weekend, without leaking the current race's outcome.
- **Leakage-Safe Historical Form:** Metrics like the driver's 3-race rolling average finish (`driver_form_3`), cumulative podiums, and the constructor's 5-race rolling form are calculated strictly from past chronologically sorted races using `.shift(1)`.
- **Context & Reliability:** Includes contextual variables like circuit altitude (`alt`), driver age, and teammate rivalry (`beat_teammate_quali`). It also uses a 10-race rolling constructor mechanical DNF rate, tracking specific historical mechanical failures such as Engine, Gearbox, and Hydraulics issues [**?**].

Known Limitations

- **Historical Qualifying-Coverage Drift:** Because modern knockout qualifying (Q1, Q2, Q3) formats did not exist in older eras, legacy data often lacks these granular lap times. Using median imputation for these missing historical values compresses the variance in single-lap pace across older seasons.
- **Era Sensitivity (Non-Stationarity):** Formula 1 is highly sensitive to major technical regulation changes, such as the shift to the ground-effect era in 2022–2024 or the historical front-engine era of 1950–1957 [**?**]. Models heavily reliant on historical constructor form struggle to adapt immediately when these regulation resets abruptly alter the competitive hierarchy.
- **Absence of Sunday Race-Day Context:** By strictly respecting the pre-race cutoff, the model is inherently blind to stochastic Sunday variables such as first-lap collisions, safety car deployments, pit-stop disasters, and sudden weather changes.